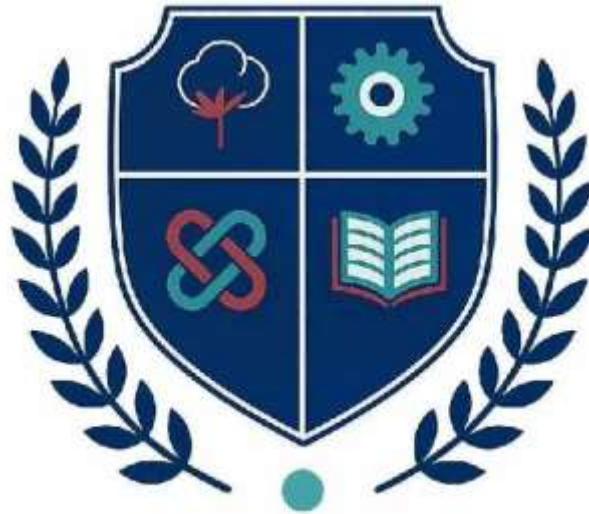






University of Kamalia

PROJECT NAME:

Student Management System

COURSE:

Object Oriented Programming (OOP)

SUBMITTED BY:

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SUBMITTED TO:

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INTRODUCTION

I have developed a simple Student Management System in C++ using Object Oriented Programming (OOP) concepts.

This system is used to store student information like name, age, roll number, department , semester and marks.

It calculates student grade and fee. The program is menu based which help user to perform different tasks easily step by step.

OBJECTIVE OF PROJECT

The main objectives of this project are as follows:

- To manage student data efficiently
- To learn OOP concepts
- To use classes and objects
- To use inheritance and encapsulation
- To use constructors and functions overloading

Classes Used

Here two classes are used:

- One is **PERSON** class that act as PARENT class.
- It contains name and age.
- And the Second is **STUDENT** class that acts as DERIVED class.
- It contains Roll No, Department, Semester and Marks.

- Student class inherits properties and characteristics of Person (Base) class using single inheritance.

OOP Concepts Used

The basic OOP concepts used in this project are as follows:

- Class(Person and Student) and Object(Student s)
- Inheritance(Single level)
- Encapsulation(private, protected)
- Constructor(default, parameterized)
- Destructor (Default)
- Constructor Overloading
- Function Overloading

Working

The program starts and shows a menu. Then user enters a choice. The system performs the selected task like adding student, showing Student information, calculating grade, and calculating fee. The program ends when user selects Exit option.

Features

This system can add student record, display student information, calculate student grade, and calculate fee. The program is simple and easy to use and understand.

Sample Output

When the program runs, a menu is shown. The user chooses an option according to his choice. The system then adds student data, shows student information, calculates grade or calculates fee. The program keeps running until Exit option is selected.

Conclusion

This project helped me understand basic OOP concepts. I learned how to use classes, objects, inheritance, constructors and function overloading. This project is simple and meets all requirements and clear all basic concepts.

Learning Outcomes

After this project, I learned OOP concepts in C++, how to use classes and objects, inheritance, constructors and destructors, function overloading. I learned how to make a simple menu based program using do while loop and switch condition.

THE END